The Assisted Climb: Technical Discovery & Implementation Plan

Complete Project Documentation

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Prepared For: Dawn Taylor, Challenge Innovate Grow
Prepared By: CloudiQS Technical Team
Project: AI Meta-Literacy Platform for Schools

Executive Summary

The Assisted Climb is an innovative AI Meta-Literacy educational framework designed to help learners "stay in the pilot seat" when using AI tools. This document outlines the comprehensive technical discovery findings and provides a detailed implementation plan to transform the proven educational concept into a secure, scalable SaaS platform for schools.

Key Highlights

- **Proven Educational Value**: Research-validated curriculum with market traction
- **Clear Technical Path**: Orchestration platform integrating with existing school AI services
- **Timeline**: 16-20 weeks to MVP (Recommended: November 2024 launch)
- **Target**: 3-5 pilot schools for initial deployment

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1. Project Overview

1.1 The Assisted Climb Framework

Mission: "Helping Learners Stay in the Pilot Seat When Using AI"

Core Philosophy:

- AI changes the conditions of learning
- Without metacognitive frameworks, learners risk bypassing learning through over-reliance on AI
- Learners must develop critical thinking and judgment skills to use AI responsibly

1.2 The Three-Pillar Methodology

THE MAP 🗺️ - Plan before you begin

- **Purpose**: Strategic planning and preparation
- **Activities**:
  - Understand the task
  - Organise thinking
  - Activate prior knowledge
  - Plan the learning journey

THE TORCH 🔦 - Question as you learn

- **Purpose**: Active inquiry and exploration
- **Activities**:
  - Explore ideas with AI
  - Clarify confusion
  - Deepen understanding
  - Compare perspectives

THE COMPASS - Check before you decide

- **Purpose**: Verification and critical evaluation
- **Activities**:
  - Verify information
  - Evaluate reliability
  - Identify bias
  - Protect judgement

1.3 Brand Identity

ORBI: The robot character representing the framework, symbolizing AI as a helpful guide rather than a replacement for human thinking.

Target Audience:

- Schools (Primary and Secondary)
- Multi-Academy Trusts (MATs)
- Teacher Training Providers
- Educational Organizations
- Universities

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2. Current State Assessment

2.1 What Exists Today

Asset	Status	Quality
Educational Framework	Complete	High - Research-validated
Curriculum Content	Complete	High - Field-tested
Brand & Assets (Orby)	Complete	High - Recognizable identity
Replit Prototype	Functional	Medium - Proof of concept only
Market Validation	Proven	High - Schools interested
Intellectual Property	Owned	High - Self-funded, 100% ownership

2.2 What Doesn't Exist

Requirement	Status	Priority
Production-ready Platform	Missing	Critical
Development Team	Missing	Critical
Technical Infrastructure	Missing	Critical
School SSO Integration	Missing	Critical
AI API Integrations	Missing	Critical
Multi-tenant Architecture	Missing	High
Teacher Dashboard	Missing	High
Security Implementation	Missing	Critical

2.3 Key Stakeholders

Dawn Taylor - Founder, Challenge Innovate Grow

- Role: Product Owner, Educational Expert, Curriculum Designer
- Responsibilities: Pedagogy, school relationships, training delivery

CloudiQS - Technical Partner

- Role: Technology Provider, Development Team, AWS Specialist
- Responsibilities: Architecture, development, deployment, infrastructure

AWS - Platform & Funding Partner

- Role: Cloud Infrastructure Provider, Funding Programs
- Responsibilities: Credits, technical support, go-to-market assistance

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3. Technical Discovery Findings

3.1 Critical Architecture Decisions

❌ DO NOT BUILD A NEW LLM

Key Discovery: The platform should **orchestrate existing school-licensed AI services**, not replace them.

Rationale:

- Schools already invest in Microsoft Copilot for Education or Google Gemini for Education
- These services are trusted, compliant, and pre-approved by educational organizations
- Building a new LLM would be cost-prohibitive and create procurement barriers
- Student data security is maximized when prompts never leave school boundaries

✅ INTEGRATION APPROACH

The Assisted Climb platform will:

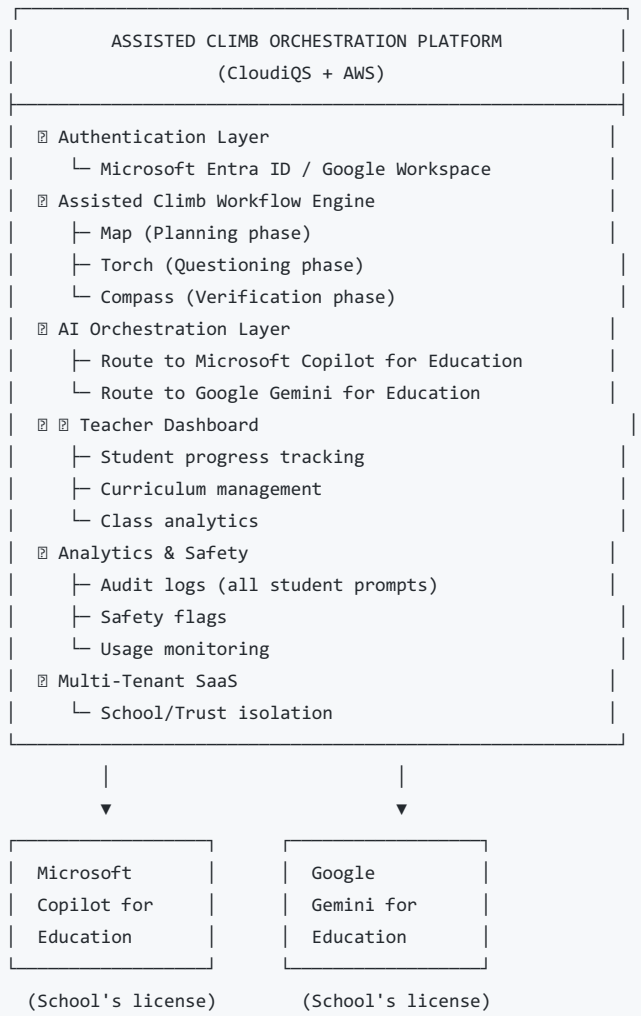
1. Sit within the school's security perimeter
2. Connect via API to the school's existing AI service (Microsoft or Google)
3. Add the Map → Torch → Compass workflow layer
4. Provide teacher oversight and student progress tracking
5. Log all interactions for audit and safety purposes

✅ SECURITY & GOVERNANCE

School Requirements:

- Student activity must be fully tracked and recorded
- Safety flags for inappropriate content
- Audit trails for compliance
- Data never leaving their educational environment
- Integration with existing school identity providers (SSO)

3.2 Platform Components



3.3 Why This Architecture is Superior

Advantage	Benefit
Schools keep existing investments	No need to replace Microsoft or Google; leverages trusted services
Enhanced security & compliance	Data stays within school boundaries; full audit trails built-in
Lower barriers to adoption	No new AI vendor to approve; uses familiar authentication
Focus on pedagogy, not infrastructure	Platform provides learning framework; schools provide AI compute
Faster time to market	No need to build/host/manage LLMs; simplified AWS architecture

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4. AWS Architecture Design

4.1 High-Level Architecture

Region: eu-west-2 (London, UK) - GDPR/UK GDPR compliance

Core AWS Services:

Service	Purpose
Amazon Cognito	User authentication, SSO federation with Microsoft/Google
AWS Lambda	AI orchestration, serverless API functions

Service	Purpose
Amazon ECS/Fargate	Container hosting for web application
Amazon RDS PostgreSQL	Student progress, tenant data, user accounts
Amazon Aurora Serverless	Audit logs, AI interaction history
Amazon S3	Curriculum content, Orby assets, backups
Amazon CloudFront	CDN for global performance
Amazon API Gateway	API routing, rate limiting
AWS Secrets Manager	Tenant-specific credentials (Microsoft/Google)
Amazon CloudWatch	Logging, monitoring, alerts
AWS CloudTrail	Compliance audit trail

4.2 Security & Compliance

Data Encryption:

- **In Transit**: TLS 1.3 (ALB, CloudFront)
- **At Rest (Database)**: AES-256 (RDS encryption)
- **At Rest (Storage)**: SSE-KMS (S3 bucket encryption)
- **Secrets**: KMS-encrypted (AWS Secrets Manager)

Compliance:

- ✅ GDPR / UK GDPR (Data residency in eu-west-2)
- ✅ ISO 27001 (AWS services certified)
- ✅ SOC 2 Type II (AWS infrastructure)
- ✅ UK Data Protection Act
- ✅ Child Safety (Age verification, parental consent workflows)

Multi-Tenancy:

- Row-Level Security (PostgreSQL RLS)
- Tenant ID in JWT tokens
- Separate S3 prefixes per tenant
- CloudWatch log isolation

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5. Integration Strategy

5.1 Microsoft 365 Education Integration

Authentication: Microsoft Entra ID (Azure AD) via SAML 2.0/OIDC

AI Integration: Microsoft Copilot for Education API via Microsoft Graph

Data Flow:

1. Student authenticates → Microsoft Entra ID
2. Cognito federates identity → JWT issued
3. Student asks question in Torch phase → Lambda function
4. Lambda calls Microsoft Graph API → Copilot for Education
5. Response returned → Logged in Aurora → Displayed to student

5.2 Google Workspace for Education Integration

Authentication: Google Workspace SSO via OIDC

AI Integration: Google Gemini for Education API

Data Flow:

1. Student authenticates → Google Workspace
2. Cognito federates identity → JWT issued
3. Student asks question in Torch phase → Lambda function
4. Lambda calls Google Gemini API → Gemini for Education
5. Response returned → Logged in Aurora → Displayed to student

5.3 Safety & Content Filtering

Pre-Prompt Filtering:

- AWS Comprehend for sentiment analysis
- PII detection (block personal information)
- Custom keyword filtering (profanity, violence)
- Age-appropriate content validation

Post-Response Filtering:

- AI response validation
- Harmful content detection
- Teacher alerts for flagged content

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6. Development Timeline

6.1 Timeline Options

Scenario	Duration	Team Size	Launch Target	Risk Level
Aggressive	12-14 weeks	5 developers	September 2024	High
Realistic ✅	16-20 weeks	4 developers	November 2024	Medium
Conservative	24-26 weeks	3 developers	January 2025	Low

Recommended: **18 weeks (November 2024 launch)**

6.2 Detailed 18-Week Timeline

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PHASE 1: DISCOVERY & PLANNING (Week 1-2)
├─ Product Discovery Workshop (3 hours)
├─ Technical Architecture Design
├─ AWS Funding Applications
└─ Development Team Assembly

PHASE 2: MVP DEVELOPMENT (Week 3-17)
├─ Sprint 1 (Week 3-5): Infrastructure + Authentication
│   ├─ AWS infrastructure setup
│   ├─ Cognito configuration
│   ├─ Microsoft/Google SSO integration
│   └─ Database schema creation
│
├─ Sprint 2 (Week 6-8): Map Phase + AI Orchestration
│   ├─ Map interface (Orby UI)
│   ├─ Lambda: Microsoft Copilot orchestrator
│   ├─ Lambda: Google Gemini orchestrator
│   └─ Safety & content filtering
│
├─ Sprint 3 (Week 9-11): Torch + Compass Phases
│   ├─ Torch interface (AI questioning)
│   ├─ Compass interface (verification)
│   ├─ Full student workflow (Map → Torch → Compass)
│   └─ Audit logging
│
├─ Sprint 4 (Week 12-14): Teacher Dashboard
│   ├─ Student list view
│   ├─ Progress tracking
│   ├─ AI interaction history
│   └─ Safety flag alerts
│
└─ Sprint 5 (Week 15-17): Multi-Tenancy + Admin
    ├─ School onboarding workflow
    ├─ Tenant isolation (RLS)
    ├─ User management
    └─ Admin console

PHASE 3: TESTING & PILOT PREP (Week 18-19)
├─ Functional testing
├─ Security testing (OWASP Top 10)
├─ Performance testing (load testing)
├─ Accessibility testing (WCAG 2.1 AA)
└─ Pilot school setup

PHASE 4: PILOT LAUNCH (Week 20)
└─ Go Live: 3-5 pilot schools
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6.3 Development Effort by Feature

Feature	Duration	Complexity	Risk
AWS Infrastructure Setup	1-2 weeks	Medium	Low
Authentication (SSO)	2-3 weeks	High	High
AI Orchestration	2-3 weeks	High	High
Student Workflow (Map/Torch/Compass)	4-5 weeks	Medium	Medium
Teacher Dashboard	2-3 weeks	Medium	Low
Multi-Tenancy	2-3 weeks	High	Medium

Feature	Duration	Complexity	Risk
Admin Console	1-2 weeks	Low	Low
Testing & QA	2-3 weeks	Medium	Medium

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7. Resource Requirements

7.1 Development Team

Recommended Team Structure (4-5 people):

Role	Allocation	Weeks	Key Responsibilities
Solutions Architect	50%	1-20	Architecture design, technical leadership
Full-Stack Developer	100%	3-17	Frontend (React) + Backend (Node.js)
Backend Developer	100%	3-17	AI integrations, API development
DevOps Engineer	100%	3-20	AWS infrastructure, CI/CD, deployment
QA Engineer	100%	15-19	Testing, security validation
UI/UX Designer	50%	3-12	Interface design, Orby integration
Product Manager	25%	1-20	Coordination, stakeholder management

7.2 Technology Stack

Frontend:

- React.js (UI framework)
- Next.js (SSR, routing)
- TypeScript (type safety)
- Material-UI or Chakra UI (component library)
- Orby character assets integration

Backend:

- Node.js (API server)
- Python (Lambda functions for AI orchestration)
- Express.js (REST API framework)
- GraphQL (optional, for complex queries)

Database:

- PostgreSQL (RDS Multi-AZ)
- Aurora Serverless (audit logs)
- Redis (ElastiCache for sessions)

DevOps & Infrastructure:

- AWS CDK or Terraform (Infrastructure as Code)
- GitHub Actions (CI/CD pipeline)
- Docker (containerization)
- ECS Fargate (container orchestration)

Security:

- AWS WAF (web application firewall)
- AWS Shield (DDoS protection)
- AWS Secrets Manager (credentials)
- AWS KMS (encryption keys)

Monitoring:

- CloudWatch (logging, metrics)
- CloudTrail (audit logs)
- X-Ray (distributed tracing)

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8. Risk Assessment

9.1 Technical Risks

Risk	Probability	Impact	Mitigation Strategy
Microsoft/Google API access issues	Low	High	Validate API access in Week 1 with pilot schools; have both integrations ready
SSO integration complexity	Medium	High	Allocate 3 weeks; use AWS Cognito best practices; test with real school tenants early
September deadline missed	High	Medium	Target November instead; strict MVP scope; weekly progress reviews
Development delays	Medium	Medium	2-week buffer built into timeline; daily standups; clear blockers early
Security vulnerabilities	Low	High	Security audit before launch; AWS best practices; penetration testing
AWS funding not approved	Low	Medium	Apply to multiple programs; have backup budget plan

9.2 Business Risks

Risk	Probability	Impact	Mitigation Strategy
Pilot schools drop out	Low	Medium	Have 5+ schools lined up; maintain regular communication; provide excellent support
Scope creep	High	High	Feature freeze after Week 2; strict change control; defer non-critical features to v2.0
Market timing (schools' budget cycles)	Medium	Medium	Align launch with academic year; understand procurement timelines
Competition emerges	Medium	Medium	Speed to market; strong pedagogy differentiator; school relationships
Regulatory changes	Low	Medium	Monitor UK education policy; maintain compliance documentation

9.3 Operational Risks

Risk	Probability	Impact	Mitigation Strategy
Key developer unavailability	Low	High	Cross-train team members; document architecture decisions; maintain code quality
CloudiQS capacity constraints	Low	High	Confirm team availability upfront; have backup developers identified
Dawn's availability for decisions	Medium	Medium	Establish decision-making framework; weekly check-ins; clear escalation path
Pilot school support burden	Medium	Medium	Document common issues; create self-service resources; teacher training program

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9. Next Steps

9.1 Immediate Actions (This Week)

CloudiQS:

- [ ] Assemble 4-5 person development team (confirm availability)
- [ ] Prepare NDA for technical deep-dive meeting
- [ ] Draft AWS Activate and EdStart funding applications
- [ ] Confirm AWS Solutions Architect availability

Dawn:

- [ ] Share Replit prototype access credentials
- [ ] Provide list of 5+ pilot school contacts

- ☐ Gather Orby brand assets (high-res graphics, style guide)
- ☐ Confirm budget approval for £50,000 - £70,000 net investment

Joint:

- ☐ Schedule 3-hour Product Discovery Workshop
- ☐ Sign mutual NDA
- ☐ Agree on communication cadence (weekly status calls)

9.2 Week 1-2 Deliverables

Product Discovery Workshop (3 hours):

- ☐ Dawn demonstrates Replit prototype
- ☐ Define student user journey (Map → Torch → Compass)
- ☐ Define teacher user journey (dashboard, monitoring)
- ☐ Identify MVP feature set
- ☐ Document technical constraints
- ☐ Agree on success metrics for pilot

Technical Architecture:

- ☐ AWS architecture diagram finalized
- ☐ Database schema (ERD) created
- ☐ API specification documented (OpenAPI/Swagger)
- ☐ Security design document
- ☐ Tech stack recommendations

AWS Funding:

- ☐ AWS Activate application submitted
- ☐ AWS EdStart application submitted
- ☐ AWS AI/ML credits application submitted
- ☐ Estimated credit allocation determined

Contracts & Agreements:

- ☐ CloudiQS engagement agreement signed
- ☐ Statement of Work (SOW) finalized
- ☐ Payment terms agreed
- ☐ IP ownership clarified

9.3 Success Criteria for Pilot Launch

Technical:

- ☐ 99% uptime during pilot period
- ☐ <2 second page load times
- ☐ Zero critical security incidents
- ☐ All pilot schools successfully onboarded
- ☐ SSO authentication working for both Microsoft and Google

Educational:

- ☐ 80%+ student engagement (complete full Map → Torch → Compass workflow)
- ☐ Positive teacher feedback (4/5 average rating)
- ☐ Measurable learning outcomes improvement
- ☐ Students demonstrate metacognitive skills in Compass phase

Business:

- ☐ 3-5 pilot schools actively using platform
- ☐ 300-500 students participating
- ☐ Case studies and testimonials collected
- ☐ Platform ready for scaling to 50+ schools

Appendix A: Delivery Pathways

The Assisted Climb offers flexible implementation models:

A.1 Whole-School Days

- Immersive collapsed timetable workshops
- AI learning experiences for entire school
- Full-day engagement with Orby and the framework

A.2 Student Programmes

- Structured AI Meta-Literacy lessons
- Guided classroom activities
- Progressive skill development through Map → Torch → Compass

A.3 Teacher Professional Learning

- AI-aware teaching workshops
- Learning and assessment support
- Integration strategies for existing curriculum

A.4 Online & Hybrid Delivery

- Live online workshops
- Virtual support sessions
- Flexible implementation options
- Remote teacher training

A.5 Modular Implementation

- Focus on specific areas:
  - Metacognition development
  - Verification skills
  - AI literacy fundamentals
- Customizable to school priorities

A.6 Bespoke Support

- Tailored programmes designed around:
  - School context
  - Curriculum priorities
  - Specific learning goals
  - Available resources

Appendix B: Competitive Advantages

B.1 Why The Assisted Climb is Different

Typical EdTech AI Product	The Assisted Climb
Generic AI chatbot	Pedagogically-grounded learning framework
One-size-fits-all access	Structured Map → Torch → Compass methodology
New AI vendor to approve	Uses school's existing licensed AI (Microsoft/Google)
Data privacy concerns	Data stays within school's security boundary
No curriculum	Research-validated curriculum + teacher training
Unknown brand	Orby character + recognizable brand identity
Software only	Complete learning experience + professional development

B.2 Key Differentiators

1. **Pedagogy-First Approach**
   - Built by educators, for educators
   - Research-validated methodology
   - Focus on metacognitive skill development
   - Not just technology, but a complete learning framework
2. **Human Agency Focus**
   - "Keeping learners in the pilot seat"
   - Prevents over-reliance on AI
   - Develops critical thinking and judgment
   - Empowers students, doesn't replace their thinking
3. **Integration Over Replacement**

3. **Integration over Replacement**
    - Works with existing school AI investments
    - No vendor lock-in
    - Supports both Microsoft and Google
    - Enhances, doesn't compete with, current tools
4. **Security & Compliance First**
    - Data never leaves school boundaries
    - Full audit trails built-in
    - GDPR/UK GDPR compliant by design
    - Child safety features integrated
5. **Proven Track Record**
    - Field-tested curriculum
    - Positive school feedback
    - Established brand (Orby)
    - Self-funded validation (no external pressure)

Appendix C: Technical Specifications Summary

C.1 Platform Requirements

Performance:

- Page load time: <2 seconds
- API response time: <500ms (95th percentile)
- Uptime: 99.9% availability
- Concurrent users: Support 1,000+ simultaneous sessions

Security:

- TLS 1.3 encryption for all connections
- AES-256 encryption at rest
- Multi-factor authentication support
- Regular security audits and penetration testing

Scalability:

- Auto-scaling to handle usage spikes
- Multi-region deployment capability
- Database read replicas for performance
- CDN for global content delivery